

Part II: Breaking Down User Stories

Smart Plant Care Device | Written by Brian McCarthy

Purpose: Break two Smart Plant Care Device user stories into theme, initiative, epic, scenarios, and actionable implementation tasks.

User Story 1: Automatic Plant Watering

Level	Definition / Completed Response
Theme	Seamless and Smart Plant Care: make plant care easier, more reliable, and less dependent on manual effort.
Initiative	Enable automatic watering for consistent plant hydration using real-time soil moisture data, plant type, and environmental conditions.
Epic	Automatic Watering System: build device capabilities that detect moisture levels, calculate watering needs, activate the watering mechanism, and notify users of watering status or reservoir issues.
User Story 1	As a plant owner, I want the device to automatically water my plant based on soil moisture and plant-specific needs so that my plant receives the right amount of water even when I am busy or away.
Scenarios	Happy path: Given the soil moisture level is below the plant-specific threshold and the reservoir contains water, when the device checks the plant condition, then it dispenses the correct amount of water and logs the watering event in the app. Edge case: Given the soil moisture level is within the healthy range, when the scheduled check occurs, then the device skips watering and shows the plant as sufficiently hydrated. Failure case: Given the soil is dry but the reservoir is empty, when the device attempts to water the plant, then it does not activate the pump and sends a mobile alert telling the user to refill the reservoir.
Tasks	1. Integrate and calibrate the soil moisture sensor for multiple plant and soil types. 2. Define moisture thresholds and watering rules based on plant profiles. 3. Develop the watering algorithm to calculate volume and timing. 4. Implement the pump or valve mechanism for controlled water delivery. 5. Build app notifications for watering completed, skipped watering, and empty reservoir status. 6. Test watering accuracy under dry, normal, humid, and reservoir-empty conditions.

User Story 2: Plant Health Monitoring

Level	Definition / Completed Response
Theme	Proactive Plant Health Management: help users identify issues early and maintain healthy growing conditions.
Initiative	Deliver actionable plant health insights by monitoring soil moisture, temperature, sunlight exposure, and sensor status in real time.
Epic	Real-Time Plant Health Monitoring: collect sensor data, display health metrics, generate alerts, recommend corrective actions, and detect sensor malfunctions.
User Story 2	As a plant owner, I want to view real-time plant health metrics and receive care recommendations so that I can quickly correct problems and keep my plant thriving.

Level	Definition / Completed Response
Scenarios	Happy path: Given soil moisture, temperature, and sunlight levels are all within the target range, when the user opens the app, then the app displays a healthy status and confirms no action is needed. Edge case: Given sunlight exposure is below the recommended threshold, when the device syncs data with the app, then the app recommends moving the plant to a brighter location or adjusting light exposure. Failure case: Given a sensor stops sending valid readings, when the system validates sensor data, then the app displays a sensor malfunction alert with troubleshooting instructions.
Tasks	1. Integrate soil moisture, temperature, and sunlight sensors into the device. 2. Build the data pipeline to transmit readings from the device to the mobile app. 3. Design the app dashboard for plant health status, trends, and alerts. 4. Develop alert rules for low light, temperature extremes, poor moisture conditions, and missing sensor data. 5. Create a care recommendation knowledge base by plant type. 6. Test sensor accuracy, app usability, alert timing, and recommendation relevance across multiple plant-care scenarios.

Evaluator Checklist Coverage

- Two complete user stories are provided.
- Each user story includes a Plant Care theme, initiative, epic, scenario details, and at least three tasks.
- Scenarios include specific contexts and expected outcomes using Given/When/Then language.